Which ONE of the following is an ADVANTAGE of circuit switching compared to packet switching?

Better fault tolerance

Can support more users for the same available capacity

Can reserve resources for a flow of packets

Lower flow setup delay

Which of the following are types of switching? Select all that apply.

Virtual circuit switching

Backpressure switching

Datagram switching

Envelope switching

What does an Ethernet switched network form, to enable efficient switching?

Minimum spanning tree

Two-connected graph

Tree rooted at min-ID bridge

Hamilton cycle

Ethernet bridges learn the path to a host from ________.

source address of a frame

destination address of a frame

master broadcast

manufacturer's NIC entry

Which of the following is/are advantages of packet switching compared to circuit switching? Select all that apply.

Solves hidden terminal problem

Statistical multiplexing

Fault-tolerance

Low header overhead

For which of the following values of peak-to-average ratio of demand, is packet switching better than circuit switching? Select all that apply.

0.1

0.5

1

10

Consider an ethernet LAN with 5 bridges, starting with all forwarding tables empty. Node P transmits a frame to node Q, and node Q replies with a frame to P. How many bridges learn the forwarding table entry for P ?

None

Exactly 1

Exactly 5

Depends on the exact connection topology

None of the other options

Consider an ethernet LAN with 5 bridges, starting with all forwarding tables empty. Node P transmits a frame to node Q, and node Q replies with a frame to P. How many bridges learn the forwarding table entry for Q ?

None

Exactly 2

Exactly 5

Depends on the exact connection topology

None of the other options

1 2 3 4

In a network where propagation delay dominates over transmission delay, which ONE of the following switching schemes has the highest delay for the first data packet?

Circuit switching

Datagram switching

Source routing

Patch routing

1 2 3 4

Which of the following is/are true about Ethernet switching? Select all that apply

It constructs a minimum cost tree in the network

It is resilient to switch or link failures

Everything within a two-hop neighbourhood is broadcast

There is a link layer acknowledgement using the sliding window protocol

What does a learning bridge learn?

IP address of the root bridge

IDs of nodes connecting to neighbouring non-Ethernet networks

Outgoing ports to reach specific hosts

Number of ports of each neighbouring bridge

In which ONE of the following schemes is resource reservation possible?

Source routing

Pure datagram switching

Virtual Circuit Switching

Hypercube switching

Which among the following is true of forwarding by routers? Select all that apply.

There are as many forwarding table entries as there are class C networks

The size of each forwarding table entry is proportional to the distance to the destination

Multiple forwarding table entries could match for a particular destination address, in which case the longest prefix match is chosen

None of the other options

Which among the following is true of IP packet format? Select all that apply.

The header length field specifies the header size in bytes

The total length field specifies the length of IP datagram including header, in bytes

The very first field is the protocol version

TTL field helps in reassembly of fragmented packets

Which among the following is true of IP addressing and forwarding? Select all that apply.

There are IP address ranges which can be reused by different organizations without being globally unique

Class A addresses end with a 1 in the first octet

Class D addresses are used for multicast

There are 16 classes of IP addresses

If an organization has 500 hosts, what is the prefix length of the network address it should ideally be assigned?

ANSWER:

23.0 - 23.0

Which among the following is true of forwarding by routers? Select all that apply.

There are as many forwarding table entries as there are class D networks

The forwarding table is different from the routing table

If multiple forwarding table entries match for a particular destination address, any one entry is chosen at random

None of the other options

1 2 3 4

Which among the following is true of IP packet format? Select all that apply.

The checksum is calculated across the entire contents include the header

The header length is 20 bytes without any header options

There is exactly one address field of length 32 bits

TTL field helps avoid packets looping forever

Which among the following is true of IP addressing and forwarding? Select all that apply.

The original class-based IP addressing had 5 address classes

The current class-less addressing allows networks to have exactly a 24-bit network address

Class A addresses are used for multicast

Class E addresses are used to broadcast to the entire Internet

How many host addresses can be accommodated in a class B network? Be sure to exclude commonly avoided host address patterns.

ANSWER:

65534.0 - 65534.0